

Asthma — The Asthma ER & Special Populations Ward (Treatment v2)



1. Reassess every 20 min

WHAT YOU SEE

An overhead light/clock '20 min' over the ER bed.

WHAT IT ENCODES

Acute exacerbations are managed with repeat SABA dosing every 20 minutes in the first hour and frequent reassessment (symptoms, work of breathing, SpO2, PEF/FEV1). Targets: SpO2 93–95%, PEF rising toward >60–80% predicted. Response in the first hour predicts disposition.

BOARD TRAP

Don't anchor on the initial severity — serial reassessment of the RESPONSE (especially PEF and CO2 trend) drives decisions; a patient who is not improving by 1 hour needs escalation, not more of the same.

2. SABA + ipratropium for severe

WHAT YOU SEE

Nebulizer masks labeled with SABA and ipratropium 'for severe'.

WHAT IT ENCODES

Acute bundle = controlled oxygen + repeated/continuous inhaled SABA (albuterol) + systemic corticosteroids early. Add ipratropium (SAMA) for severe/life-threatening attacks or poor initial response; combined SABA+SAMA reduces admissions. Consider IV magnesium sulfate (2 g) for severe attacks not responding to initial therapy.

BOARD TRAP

Ipratropium is add-on for SEVERE exacerbations and the acute ER setting only — it has no role in chronic maintenance asthma (that's tiotropium, a LAMA).

3. Danger signs panel

WHAT YOU SEE

A red 'DANGER SIGNS: silent chest, impending failure' panel.

WHAT IT ENCODES

Life-threatening features: silent chest (no air movement = airflow too low to wheeze), cyanosis, bradycardia/hypotension, exhaustion, confusion/altered mental status, and feeble respiratory effort. These mandate ICU and intubation readiness.

BOARD TRAP

A 'quiet' chest in a distressed asthmatic is ominous, NOT reassuring — disappearance of wheeze means worsening obstruction, not improvement.

4. Normal CO2 = ominous

WHAT YOU SEE

A gauge 'normal CO2 = ominous' with '↑ mmHg lag'.

WHAT IT ENCODES

Early in an attack tachypnea drives PaCO2 DOWN (respiratory alkalosis). A 'normalizing' or rising PaCO2 means the patient can no longer maintain the high minute ventilation — fatigue and impending respiratory failure. A normal PaCO2 in a tachypneic severe asthmatic is a red flag.

BOARD TRAP

Don't be reassured by a 'normal' PCO2 on the ABG of a tiring asthmatic — it signals failure; the only reassuring PCO2 early is a low one.

5. Exhaustion / one-word speech

WHAT YOU SEE

'Exhaustion → 1-word speech' note in danger box.

WHAT IT ENCODES

Speech fragmentation grades severity: full sentences = mild; phrases = moderate; single words / unable to speak = severe/life-threatening. Combined with accessory muscle use and inability to lie flat, it reflects critical airflow limitation.

BOARD TRAP

One-word speech plus exhaustion is a severe/life-threatening category — these patients need aggressive therapy and ICU, not discharge after a single nebulizer.

6. Ventilator: low rate, permissive hypercapnia

WHAT YOU SEE

A 'vent settings: low rate, permissive hypercapnia' card.

WHAT IT ENCODES

If intubated, set LOW respiratory rate, low tidal volume, and a LONG expiratory time (low I:E ratio) to allow full exhalation and avoid dynamic hyperinflation (breath-stacking/auto-PEEP). Tolerate a high PaCO2 and low pH (permissive hypercapnia) to prevent barotrauma.

BOARD TRAP

Hyperventilating the intubated asthmatic to 'normalize' CO2 causes air-trapping → auto-PEEP → hypotension and pneumothorax. If the patient suddenly decompensates on the vent, disconnect the circuit to let air escape and check for tension pneumothorax.

7. AERD bed (COX-2/ASA-safer)

WHAT YOU SEE

A patient labeled 'AERD: asthma + nasal polyps + NSAID reaction; COX-2 / ASA safer'.

WHAT IT ENCODES

AERD / Samter's triad = asthma + chronic rhinosinusitis with nasal polyps + reaction to aspirin/NSAIDs. Mechanism: COX-1 inhibition shunts arachidonic acid to the 5-lipoxygenase pathway → cysteinyl leukotriene overproduction → bronchospasm, rhinorrhea, flushing. Leukotriene-modifiers (montelukast, zileuton) help; definitive option is aspirin desensitization.

BOARD TRAP

Avoid ALL nonselective COX-1 NSACs (aspirin, ibuprofen, naproxen) in AERD. Selective COX-2 inhibitors (celecoxib) and acetaminophen are generally tolerated. Mnemonic: Samter = aSthma + Salicylate sensitivity + Sinus polyps.

8. EIB bed (warm-up, pre-SABA)

WHAT YOU SEE

A patient labeled 'EIB: warm up, pre-SABA, LTRA, tachyphylaxis'.

WHAT IT ENCODES

Exercise-induced bronchoconstriction: transient airway narrowing from water/heat loss during exercise (worse in cold, dry air). Prevention: pre-exercise SABA 10–15 min before activity, an adequate warm-up period, and LTRA or daily ICS for frequent symptoms. EIB often signals poorly controlled underlying asthma.

BOARD TRAP

Relying on daily pre-exercise SABA causes tachyphylaxis (tolerance, reduced bronchoprotection) — frequent EIB should be controlled with daily ICS, and well-controlled underlying asthma minimizes EIB.

9. Immunotherapy 'start' note

WHAT YOU SEE

An 'immunotherapy: start' panel with fluticasone/beclomethasone note.

WHAT IT ENCODES

Allergen immunotherapy (SCIT/SLIT) can be considered for allergic asthma driven by a specific aeroallergen when asthma is controlled — it modifies the allergic response over years. It is contraindicated in poorly controlled/severe asthma due to anaphylaxis risk.

BOARD TRAP

Never start immunotherapy in uncontrolled or severe asthma — control the asthma first; uncontrolled asthma is a contraindication because of the risk of fatal anaphylaxis.

10. Pregnancy bed (don't stop ICS)

WHAT YOU SEE

A pregnant patient labeled 'uncontrolled asthma worse than meds'.

WHAT IT ENCODES

Continue all asthma controllers in pregnancy — uncontrolled asthma (hypoxia) harms the fetus far more than the medications. Budesonide is the best-studied/preferred ICS, albuterol the preferred SABA; LABAs and oral steroids (for severe disease) are continued when needed.

BOARD TRAP

Do NOT down-titrate or stop ICS/SABA in pregnancy out of fear of teratogenicity — the maternal-fetal risk of an uncontrolled attack and hypoxia dominates. Same principle in lactation: keep treating.

11. ACO bed (don't omit ICS, watch beta-agonist)

WHAT YOU SEE

An older patient 'don't omit ICS — watch beta-agonist arrhythmia'.

WHAT IT ENCODES

Asthma-COPD overlap has features of both. Unlike pure COPD (where ICS is selective and LABA/LAMA can be the backbone), the asthma component mandates an ICS to prevent exacerbations/death — so ACO is treated ICS-LABA ± LAMA. Beta-agonists can precipitate tachyarrhythmias and hypokalemia in older patients.

BOARD TRAP

Never treat ACO with LABA or LABA/LAMA WITHOUT ICS — that's the COPD reflex, but the asthma component makes LABA-without-ICS dangerous. Watch for beta-agonist-induced arrhythmia/hypokalemia in the elderly.

12. Steroids lag → bronchodilator first; avoid late intubation

WHAT YOU SEE

A doctor with a 'steroids lag → bronchodilator first; intubation avoid hyperventilation' scroll.

WHAT IT ENCODES

Systemic corticosteroids are given EARLY (within the first hour) but take 4–6 hours to act, so immediate relief comes from inhaled bronchodilators. Plan ventilation strategy ahead to avoid air-trapping, and don't delay intubation in a fatiguing patient — but intubate semi-electively rather than as a crash.

BOARD TRAP

Steroids do not relieve acute bronchospasm in the first hours — still give them early (they prevent relapse/late deterioration), but rely on SABA ± ipratropium ± magnesium for immediate bronchodilation.

13. Home plan — ICS + steroid burst

WHAT YOU SEE

A 'HOME PLAN: ICS ×4-5 for 7-14 days' card by the bedside.

WHAT IT ENCODES

Discharge after an exacerbation: prescribe a short course of oral corticosteroids (prednisone ~40–50 mg for 5–7 days, no taper needed for short courses), START or step UP inhaled ICS (often quadrupling the maintenance dose transiently), ensure the patient has a reliever and a written action plan, check technique, and arrange follow-up within days.

BOARD TRAP

Discharging on a SABA alone after an ER visit is wrong — every exacerbation requires ICS initiation/escalation plus an oral steroid course, because the underlying inflammation persists for weeks.